

Attachment 1: Helicopter Autorotational/ High Risk Testing Trainer (HATT), Initial Requirements

Parameter	Threshold (Required)	Objective (Desired)
General		
Cockpit	Two-place / Dual controls Environmental control with Heating/Ventilation/Air Conditioning (HVAC) and /or removable doors in place of air condition	-
Rotor System	Conventional Helicopter, Main rotor / tail rotor, tandem, or counter rotating.	-
Flight Control System	Fully safe and effective unaugment controls or hydraulic boost controls with appropriate Stability Augmentation System/ Stability and Control Augmentation System (SAS /SCAS) augmentation. Trim system with FTR (Force Trim Release) switch and/or incremental trim.	Control system which provides for: 1. AFCS degradable to unaugment control in all flight regimes. 2. Auto Pilot with flight condition capture and hold modes.
External Stores / Cargo ballast	No	1. Ability to mount weapons or external fuel tanks. 2. Cargo passenger compartment suitable for ballast within the allowable Center of Gravity (CG) limits
Engine - Rotor Governing System / Engine Control	Easily manipulated throttles for throttle idle and power recovery autorotations. Engine-rotor response to permit safe, effective, repeated power recovery to wave off throughout descent and flare while remaining within the established limits.	-Full Authority Digital Engine Control (FADEC) control easily manipulated, safe, and effective for in flight recovery from idle to powered flight Direct control of variable Nr for performance measurement testing. -Capable of in-flight engine shutdown and restart
Ground servicing/turnaround	-No unique or aircraft-specific ground power or ground support equipment required	Hot-pit refueling with pilot hot-switch capability
Maintenance / Supply	Interim maintenance support from the OEM while we transition to either contracted or organic support (24 Month Period) Parts and materials available for at least two of the first phase events and an option for parts available for the first major phase event.	Minimum of 95% material availability for the most commonly used components and order fulfillment within 1 week of placing the order

Performance		
Start / Shutdown/ Rotor Engagement	-	-45 knots any azimuth -Rotor Brake
Mission Length	2 Hours	2.5 Hours
Max. Airspeed, knots	-	150 forward 45 lateral / aft
Hover / Vertical Climb Capability	Out of Ground Effect (OGE) hover on MIL STD hot day, Sea Level mid mission GW	OGE hover and 500 fpm (feet per minute) vertical climb, MIL STD hot day, Sea Level Maximum T/O GW
n_z	+ .5g to +2.0g	0 to 3.0g
Aircraft Attitude and rate Limits	30 deg pitch 60 deg roll 30 deg/ sec yaw rate	45 deg pitch 60 sustained transient 90 deg roll 60 deg/ sec yaw rate
Autorotation	Capable of safe and effective repeatable autorotation at idle to a prepared soft or hard surface at 15 knots ground speed or less without restriction on number of repetitions per flight hour or annually / TBO (gear cycles or otherwise). Without restriction to gross weight.	Rotor inertia normalized by gross weight and disk loading to provide good autorotative characteristics to include a reasonable decay from throttle reduction. Effectiveness and behavior similar to OH-58, UH-1B, AH-1S.

Safety		
Seats / Crash Attenuation	Five-point restraint harnesses for all occupants, or demonstrated equivalent survivability -Jettison-able Cockpit Doors, meeting egress time requirements. - Jettison-able / removable cabin egress port. Self-sealing fuel cells	Threshold + Crash attenuating landing gear Crash Attenuating Seats
Traffic Collision Avoidance Systems (TCAS)	TCAS w/ADS-B out (Automatic Dependent Surveillance-Broadcast)	TCAS with w/ ADS-B in/out. Map display of ADS-B in traffic, weather, etc.-
Altitude Warning	Radar Altimeter with Low Height warning setting and visual and audible signal	Enhanced Ground Proximity Warning System (EGWPS) DH / Altitude Alert settings integrated w/ PFD
Flight Data Recorder	Downloadable w/TSPI, Flight Control Position, and Systems State data	

Systems		
Navigation Suite	Visual Flight Rules (VFR) with GPS, moving map and display of ground / inertial speed and track angle	Instrument Flight Rules (IFR) certified w/RNAV (to include LPV – Localizer Performance with Vertical Guidance)
Recording System	Digital cockpit audio	Digital cockpit audio & visual w/advanced debriefing capability
Instrumentation Data System	Access to bus data or discrete signals (wherever they exist) for all cockpit indications, control positions, actuator positions, aircraft state information (attitudes, rates, accelerations, etc.) Time Space and Position Information (TSPI), and navigation information. Ability and information (ICDs etc.) necessary to instrument flight test parameters where they are not available.	Integrated Flight Test Data System to capture, display and record threshold data. Low Airspeed Data System displayed and recordable. Operational “LAADS, HADS” type or flight test instrumentation.
Weapons / Sensor Systems	-	- Integrated Weapon aiming system through fixed HUD / Reticle or Helmet Mounted Sight and Display (HMSD) System